MPSYS TECHNICAL MINI FAIR

SSY226: Design Project in MPSYS - 2025

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Synthetic Data Generation for Multi-view BEV Segmentation

INTRODUCTION

Understanding industrial environments at a semantic level is key to ensuring mobile robots can operate safely and efficiently alongside humans. However, generating reliable **bird's-eye-view (BEV)** semantic ground truth from real-world data is both expensive and difficult to scale[1]. To address this, we use **Unreal Engine** and **Gaussian Splatting** to create photorealistic synthetic factory scenes with precise multi-camera semantic labels. This setup allows us to generate BEV datasets in a fully automated, scalable, and cost-effective way, and we further introduce **a new multi-view deep learning model** designed to produce BEV representations directly.

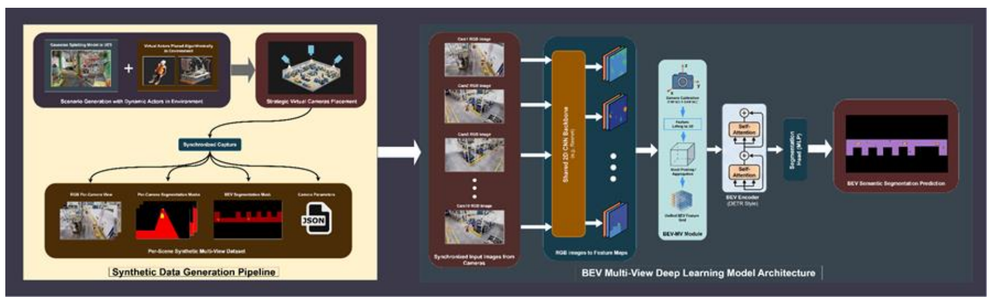


Fig.1: Overall system diagram

Highlights:

Synthetic Data Pipeline:
We have developed an automated pipeline for large-scale **synthetic** dataset generation in UE5 for a real **factory** environment. It features:

- ❑ Controlled and diverse **scene generation** using synthetic actors.
- ❑ **Synchronized** multi-camera capture from different views.
- ❑ Rendering of pixel-level **semantic masks** and **BEV** ground truth.
- ❑ Automated bundling and export for multi-scene **datasets** for model training.

BEV Semantic Model:
We have proposed an architecture, part of it inspired from BEV-SUSHI[2], that do early multi-view fusion and it has **three** main components:

- ❑ Multi-View Feature Extraction Backbone using **ResNet-50**[3].
- ❑ BEV Feature Transformation **(Spatial Cross-Attention)**[4].
- ❑ BEV Segmentation Head using **DETR** style transformer[5].

Gaussian Splatting Model:
Used a Gaussian Splatting Model[6] captured from a **real** factory for realism and novel views.

SYNTHETIC DATASET GENERATION

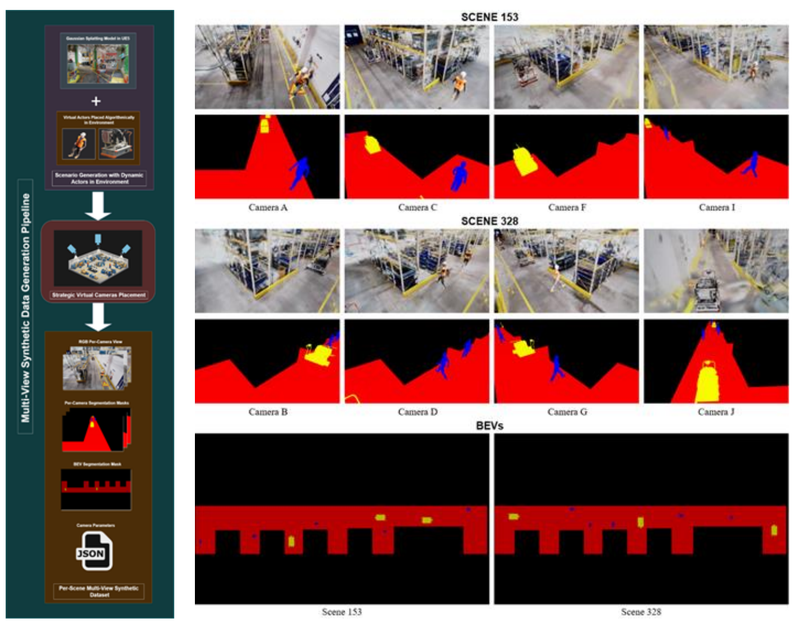


Fig.2: Synthetic multi-view dataset pipeline and ground truth images

- **Multi-view Scene Setup:**
  10 Cine-Cameras captured the scenes with 5 human and 3 robot actors, including deliberate occlusions. An overhead orthographic camera recorded the BEV.
- **Animation & Capture:**
  The Level Sequencer was used tp animate the scene for 420 frames. MRQ was executed independently for each camera to overcome plugin limitations.
- **Data Output:**
  Generated a total of 4200 images. Actor positions and orientations were saved per frame as JSON for each view.

BEV-SEMANTIC-SUSHI

Spatial Cross-Attention (View Transformer):

$$\mathbf{B} = \mathbf{Q}_{\text{BEV}} + \sum_{d=1}^{D} w_d \cdot \text{sample}\left(\mathbf{F}_{\text{cam}}, \pi(\mathbf{x}, \mathbf{y}, z_d)\right)$$

where $\mathbf{Q}_{\text{BEV}} \in \mathbb{R}^{C\times H\times W}$ are learnable BEV queries, π projects 3D points (x, y, z_d) to image coordinates using camera calibration, and w_d are learned depth weights across $D=4$ reference heights.

Deformable Cross-Attention (Mask2Former Decoder):

$$\text{DefAttn}(\mathbf{q}, \mathbf{p}) = \sum_{k=1}^{K} A_k \cdot \mathbf{V}(\mathbf{p} + \Delta\mathbf{p}_k)$$

where $\mathbf{q}$ are query features, $\mathbf{p}$ is the reference point, $\Delta\mathbf{p}_k$ are $K=4$ learned sampling offsets, and A_k are attention weights.

Total Loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{focal}} + 2\mathcal{L}_{\text{dice}} + 0.5\mathcal{L}_{\text{boundary}} + \mathcal{L}_{\text{lovász}}$$

where $\mathcal{L}_{\text{focal}}$ is weighted cross-entropy focusing on hard examples ($\gamma=2$), $\mathcal{L}_{\text{dice}}$ optimizes segmentation overlap, $\mathcal{L}_{\text{boundary}}$ penalizes edge errors, and $\mathcal{L}_{\text{lovász}}$ is a differentiable IoU surrogate.

TABLE I
TRAINING CONFIGURATION

Parameter	Value
Input image resolution	640 × 480 pixels
BEV output resolution	480 × 270 pixels
Number of cameras	10
Reference heights	4 (0.0–2.5 m)
Epochs	10
Batch size	1
Optimizer	AdamW
Learning rate	1×10^{-4}
FPN Backbone learning rate	1×10^{-5} (0.1× base)
Weight decay	0.05
Warmup epochs	5
Gradient clipping	0.5
Train/Val split	80% / 20%
Augmentation	Brightness, contrast, saturation, noise
Device	Apple MPS

RESULTS AND CONCLUSION

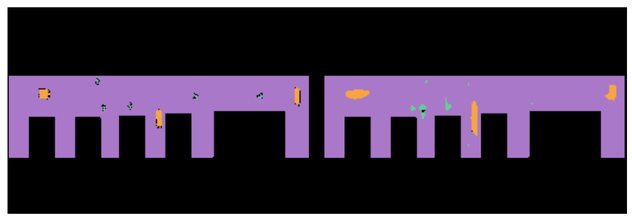


Fig.3: BEV for Ground Truth (left) and training prediction (right)



Fig.4: BEV for Ground Truth (left) and validation prediction (right)

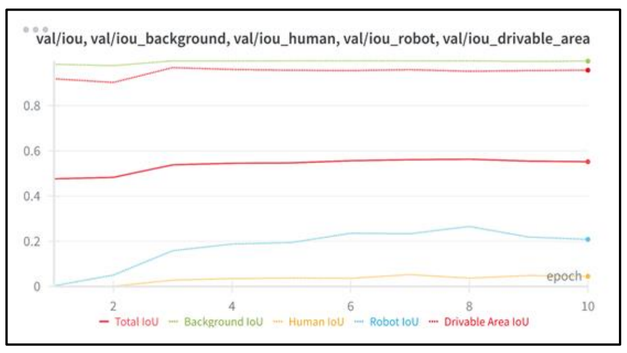


Fig.3: Training and validation losses



Fig.5: Intersection over Union (IoU)

Conclusion

- ❑ **Needs a larger dataset**
- ❑ **Better results for a class with large area**
- ❑ **Losses decreased**
- ❑ **IoU increased**

References

[1] Philion et al., Lift, Splat, Shoot: Encoding Images from Arbitrary Camera Rigs by Implicitly Unprojecting to 3D, ECCV, 2020.
[2] Wang et al., BEV-SUSHI: Multi-Target Multi-Camera 3D Detection and Tracking in Bird's-Eye View, arXiv 2024.
[3] He et al., Deep Residual Learning for Image Recognition, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016.
[4] Li et al., BEVFormer: Learning Bird's-Eye-View Representation from Multi-Camera Images via Spatiotemporal Transformers, 2022.
[5] Carion et al., End-to-End Object Detection with Transformers, Proceedings of the European Conference on Computer Vision (ECCV), 2020.
[6] Ahmed et al., Realistisk träningsdata med 3D Gaussian Splatting, 2025.